

Phase 3 - Language Definition

4 Schema and Ontologies

The cleaned datasets contain a range of entities and properties essential for analyzing oncoviruses and the interactions between viral and host proteins that disrupt cellular pathways leading to cancer. While most entity types do not have suitable representations in Schema.org or Bioschemas.org, we were able to identify corresponding concepts for many of them in NCBO BioPortal under specialized biomedical ontologies. When applicable, these entity types are aligned with the established ontology terms; otherwise, we rely on schema derived directly from the original datasets.

4.1 Cancer *(From the Human Disease Ontology, DOID:162 via NCBO BioPortal)*

- **cancerID** (Integer, from the original data): Unique identifier for the cancer entry.
- **cancerType** (Text, from the original data): Category or type of cancer.
- **primarySite** (Text, from the original data): Anatomical site or primary location associated with the cancer type.

4.2 HostProtein *(from the original data)*

- **hostUniprotID** (Text, from the original data): Unique reference to the UniProt Accession Number of the host protein.
- **hostGeneName** (Text, from the original data): Gene name of the host protein.
- **hostProteinName** (Text, from the original data): Protein name of the host protein.

4.3 ViralProtein *(from the original data)*

- **viralUniprotID** (Text, from the original data): Unique reference to the UniProt Accession Number of the viral protein.
- **viralGeneName** (Text, from the original data): Gene name encoding the viral protein.
- **viralProteinName** (Text, from the original data): Protein name of the viral protein.

4.4 ProteinAnnotation *(from the original data)*

- **proteinAnnotationID** (Text, from the original data): Unique identifier for the protein annotation entry.
- **function** (Text, from the original data): Functional role of the protein.
- **localization** (Text, from the original data): Protein's subcellular localization.

4.5 Vaccine *(From Vaccine Ontology, VO_0000001 via NCBO BioPortal)*

- **vaccineID** (Integer, from the original data): Unique identifier for the vaccine entry.

- **vaccineName** (Text, from the schema.org): Name of the vaccine.
- **vaccineDescription** (Text, from schema.org): Description of the vaccine.
- **vaccineSource** (URL, from the original data): A reference to the origin of the vaccine information, typically a webpage, database entry, report, or publication from which the vaccine's entity data was obtained.

4.6 Virus (*From NCBITAXON, STY/T005 via NCBO BioPortal*)

- **virusCode** (Text, from the original data): Unique code for the virus.
- **organismName** (Text, from the original data): Scientific or common name of the virus.
- **species** (Text, from Invasion Biology Ontology, NCIT_C45293, *via NCBO BioPortal*): Viral species name.
- **host** (Text, from the original data): Primary host organism.

4.7 Phenotype (*from bioschemas.org*)

- **phenotypeID** (Integer, from the original data): Unique identifier for the phenotype entry.
- **virusType** (Text, from the original data): Category or type of the virus.
- **phenotypeSource** (URL, from the original data): A reference to the origin of the phenotype information, typically a webpage, database entry, report, or publication from which the virus's phenotype entity data was obtained.
- **trait** (Text, from the original data): Described phenotypic trait.

4.8 Pathway (*from Pathway Ontology, PW_0000001, via NCBO BioPortal*)

- **pathwayKeggID** (Text, from the original data): Unique reference to KEGG pathway database.
- **pathwayName** (Text, from the original data): Name of the biological pathway.
- **pathwayCategory** (Text, from the Schema.org): Pathway classification.
- **pathwayDescription** (Text, from Schema.org): Description of the pathway.
- **pathwaySource** (URL, from the original data): A reference to the origin of the information, typically a webpage, database entry, report, or publication from which the pathway's entity data was obtained.

This schema captures the structure of the datasets while integrating Schema.org and Bioschemas.org standards when suitable, offering a comprehensive framework for analyzing oncoviruses and the interactions between viral proteins and host proteins that disrupt cellular pathways leading to cancer.

5 Language Definition

The primary goal of the language definition phase is to formally specify the concepts and information that will be included in the final Knowledge Graph to fulfill the objectives of this project. To achieve this, we first identified all relevant concept structures, including E-Types, relationships, and object properties, that needed formal specification. Each identified concept was then aligned with existing ontologies, first searching on the Universal

Knowledge Core (UKC) to examine whether the concept was already defined. If a concept was not present, it was subsequently checked against additional specialized resources, and if still not found, it was defined as a custom concept specific to our project. Due to the nature of this study, many of the elements are highly domain-specific, for example, the E-Type HostProtein and its properties, and therefore required a complete formal definition within our framework. Since our dataset contains numerous concepts specific to molecular biology, virology, and immunology, we relied on specialized ontologies to retrieve definitions and standard identifiers, completing our Language Resource Table. In particular, we used ontologies from NCBO BioPortal.

5.1 Concept Identification

We produced the Language Resource Table, in which each identified concept is listed and organized into three columns:

- Column 1: ConceptID: This takes one of three formats depending on where the concept was found: a UKC ID (ID: UKC-number) if present in the Universal Knowledge Core, a link to an external ontology if found elsewhere, or a custom project ID (ID: OKG25-number) if defined internally.
- Column 2: Word_en: The name or word(s) associated with the concept.
- Column 3: Gloss-en: A clear explanation or definition for each concept.

In order to improve readability, we applied the following color-coding scheme:

- **Blue:** Represents ETypes from the ER Model
- **Green:** Represents Attributes from the ER Model
- **Red:** Represents Relationships from the ER Model

ConceptID	Word-en	Gloss-en
UKC-6543	Virus	Ultramicroscopic infectious agent that replicates itself only within cells of living hosts; many are pathogenic; a piece of nucleic acid (DNA or RNA) wrapped in a thin coat of protein.
UKC-24644	Vaccine	Immunogen consisting of a suspension of weakened or dead pathogenic cells injected in order to stimulate the production of antibodies.
UKC-26778	Phenotype	What an organism looks like as a consequence of the interaction of its genotype and the environment.
UKC-67961	Cancer	Any malignant growth or tumor caused by abnormal and uncontrolled cell division; it may spread to other parts of the body through the lymphatic system or the blood stream.
http://purl.obolibrary.org/obo/COMO_0000067	ProteinAnnotation	Additional information and metadata on proteins.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C17	ViralProtein	Proteins encoded by viral genes.

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http://purl.obolibrary.org/obo/IDOMAL_0001068	HostProtein	A protein that is produced by a host.
http://www.biopax.org/release/biopax-level3.owl#Pathway	Pathway	A set or series of interactions, often forming a network, which biologists have found useful to group together for organizational, historic, biophysical or other reasons. Usage: Pathways can be used for demarcating any subnetwork of a BioPAX model. It is also possible to define a pathway without specifying the interactions within the pathway. In this case, the pathway instance could consist simply of a name and could be treated as a 'black box'. Pathways can overlap, i.e. a single interaction might belong to multiple pathways. Pathways can also contain sub-pathways. Synonyms: network, module, cascade, Examples: glycolysis, valine biosynthesis, EGFR signaling.
UKC-695	localization	A determination of the place where something is.
UKC-6810	host	An animal or plant that nourishes and supports a parasite; it does not benefit and is often harmed by the association.
UKC-105757	function	Serve a purpose, role, or function.
http://edamontology.org/data_2909	organismName	The name of an organism (or group of organisms).
http://edamontology.org/data_1045	species	The name of a species (typically a taxonomic group) of organism.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C85496	trait	Any genetically determined characteristic.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C164895	pathwayName	Name of the biological pathway.
http://edamontology.org/data_2343	pathwayKeggID	Identifier of a pathway from the KEGG pathway database.
http://ncicb.nci.nih.gov/xml/owl/EVS/Thesaurus.owl#C164893	pathwayDescription	A brief explanation of what a pathway does or represents.
http://sbmi.uth.tmc.edu/ontol	primarySite	Primary Site / Anatomical site or primary location associated with the cancer type.

ogy/ochv#C0449695		
http://edamontology.org/data_2592	cancerType	A type (represented as a string) of cancer.
http://purl.bioontology.org/ontology/SNOMEDCT/277234002	virusType	Describes the category or classification of a virus.
OKG25-01	virusCode	Unique identifier assigned to a virus.
OKG25-02	cancerID	Unique identifier for the cancer record.
OKG25-03	vaccineID	Unique identifier for the vaccine record.
OKG25-04	vaccineName	The official or common name of a vaccine.
OKG25-05	vaccineDescription	A brief explanation of what the vaccine does and what it targets or prevents.
OKG25-06	vaccineSource	The origin, database, or reference from which vaccine information is obtained.
OKG25-07	phenotypeID	Unique identifier for the virus phenotype record.
OKG25-08	phenotypeSource	The source or reference documenting the described phenotype.
OKG25-09	pathwayCategory	The broad functional class or grouping of a biological pathway.
OKG25-10	pathwaySource	The origin, database, or reference providing the pathway information.
OKG25-11	viralUniprotID	Unique Uniprot Reference to the viral protein.
OKG25-12	hostUniprotID	Unique Uniprot Reference to the host protein.
OKG25-13	viralGeneName	Standard name assigned to a viral gene.
OKG25-14	hostGeneName	Standard name assigned to a host gene.
OKG25-15	viralProteinName	Official or common name of a viral protein.
OKG25-16	hostProteinName	Official or common name of a host protein.
OKG25-17	proteinAnnotationID	Unique identifier for the protein annotation record.
UKC-98182	targets	Intend (something) to move towards a certain goal.
UKC-100755	causes	Give rise to; cause to happen or occur, not always intentionally.
OKG25-18	involvedIn	Indicates that an entity participates in, contributes to, or plays a role within a process, event, pathway, or higher-level activity. The relationship is generally directional, linking the participating entity (subject) to the broader process or context (object).
OKG25-19	interactsWith	Denotes that two entities engage in a direct or measurable interaction, influencing each other's state, function, or behavior. This relationship is typically symmetric, reflecting mutual participation in the interaction.
OKG25-20	hasAnnotation	Links an entity to a descriptive annotation or metadata.
OKG25-21	hasPhenotype	Links an entity to an observed trait, characteristic, or phenotype.

OKG25-22	encodes	Represents that a genetic element (such as a gene, transcript, or genomic region) of an organism specifies the sequence or structure of a functional product, typically a protein or RNA molecule. The relationship links the genetic source (subject) to the resulting biological product (object).
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5.2 Language Teleontology

In this step of the project, the focus was on constructing the language teleontology. Using the Protégé software, we formalized the language by defining the list of EType terms, object property terms, and data property terms as an OWL file. This included aligning our language to one or more reference language teleontologies, ensuring consistency and semantic clarity. Teleontology was then constructed to encode the domain language representation of our specific purpose, capturing object and data properties. The resulting OWL file, named [oncovirus_KG_language_teleontology.owl](#), is available on our GitHub repository. This phase establishes a solid foundation by formalizing the structure and terminology of the language, preparing it for future integration into the complete knowledge graph and the subsequent formalization of the full teleology.

5.3 Qualitative Evaluation

During the language definition phase, one relationship in the ER diagram, linking Pathway to Cancer, was removed due to inconsistencies identified in the underlying data. This adjustment did not result in any loss of meaningful information, as the association can still be inferred indirectly through host proteins and their involvement in cancer-related pathways. The informal purpose of the knowledge graph remained fully aligned with the original intent, confirming the robustness of the conceptual design. In refining the schema, several relationships and attributes were renamed to improve semantic clarity, and redundant identifiers were removed where authoritative external IDs already served as unique identifiers. Corresponding adjustments were made in the datasets to ensure naming consistency and conciseness. The evaluation also revealed the need for a small number of supplementary attributes, which were retrieved and integrated without difficulty. Collectively, these refinements strengthened the coherence and precision of the schema while maintaining full compatibility with the biological evidence and data sources.